

NoseKnowsNorth: Directional Cueing using Nose Vibration Stimulation for Smart Glasses

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Figure 1: (a) The device is mounted on XREAL Air, and when worn together, vibration stimuli are presented to the nose and direction can be recognized. (b) Device configuration. A vibration motor is built into the nose pad, and a system consisting of a microcontroller, sensor, and battery is attached to the vine.

ABSTRACT

Most directional navigation systems used in daily life provide visual and auditory information through smartphones or smart glasses. However, visual information may pose a risk of taking away the user's attention and may cause a load due to information overload. Auditory information may be difficult to hear due to ambient noise. To solve this problem, previous studies have proposed methods to provide tactile stimulation to the neck, wrists, and face. Direction presentation using the sense of touch can convey direction without interfering with vision or hearing. However, there are few devices that can be worn for long periods of time in daily life with little burden. In this study, we propose NoseKnowsNorth, a device that provides directional cueing by delivering vibrational stimuli to both sides of the nasal bridge, leveraging the nose's high tactile sensitivity. The nose can transmit information with weak vibrations, and by incorporating the device into the nose pads of eyeglasses, it can be easily used with smart glasses and reduce the burden of wearing them. The experimental results showed that left-right discrimination was achievable with nearly 100% accuracy, and

directions could be indicated with a resolution within 45° .

Index Terms: Tactile, Nose, Smart Glasses

1 INTRODUCTION

Losing a sense of direction in daily life can be inconvenient. For instance, when walking in a city and becoming unsure of the route to a destination, one might wander in the wrong direction, resulting in wasted time and physical fatigue. Such a lack of directional sense can pose life-threatening risks, especially in situations where spatial awareness is critical. For example, spatial disorientation accounts for approximately 33 % of aviation accidents, nearly 100 % of which are fatal [2]. Similarly, in scuba diving, losing a sense of direction underwater can make it difficult to ascend to the surface, leading to risks such as getting lost, oxygen depletion, or severe accidents caused by changes in water pressure.

To address these issues, map applications on smartphones or smart glasses are commonly used in daily life. These applications assist users in reaching their destinations through visual displays and audio guidance. However, visual information displays often require users to focus on the screen, which reduces attention to their surroundings and increases the risk of collisions or falls. While smart glasses can provide information without significantly obstructing the user's vision, challenges remain, such as reduced visibility due to environmental interference or user overload from excessive information. On the other hand, auditory information de-

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livery can be difficult to hear in noisy surroundings, and notification sounds interrupting the music while it is playing can diminish the quality of the listening experience.

In aviation, to mitigate spatial disorientation, systems have been introduced that provide pilots with multi-modal warnings using visual, auditory, and tactile cues, in addition to systems designed to prevent ground collisions [6]. This highlights the potential of tactile stimuli as an alternative to visual and auditory information delivery. Previous studies [1, 3, 4, 5, 7, 8, 9, 10, 11, 12] have proposed methods for directional guidance using tactile feedback. However, these methods have yet to achieve a balance where they do not interfere with vision or hearing while remaining comfortable for long-term use in daily life.

In this study, we focus on the nose as a low-burden area for wearable devices and propose a method for directional guidance utilizing the sides of the nasal bridge. We evaluate the effectiveness of tactile feedback on the nose as a means of directional guidance using a prototype device developed for this purpose.

2 RELATED WORK

2.1 Direction Indication Using Haptics

Various approaches have been researched for indicating direction and providing guidance using haptic feedback. Examples include devices worn around the neck[5, 10] or waist[12] that indicate direction through vibration, a method for controlling the orientation of the head by applying electrical muscle stimulation (EMS) to the neck muscles[11], and devices that provide directional cues by delivering thermal stimuli around the wrist[9].

In addition, belt-type devices worn around the waist [7] and skin-attached devices called NorthSense¹ aim to cultivate a sense of direction by conveying the location of the north through vibration. These methods seek to explore new perceptual experiences by offering humans a sense—orientation to north—that they do not inherently possess. However, because one typically moves their head when confirming direction, leveraging sensations on the face may provide a more intuitive means of conveying directional cues. Moreover, the face has much higher sensitivity compared to other parts of the body, making it possible to perceive even weak stimuli [14]. Therefore, this study focuses on the face as a means to achieve a more intuitive link between the act of facing a direction and haptic cues.

2.2 Sensory Presentation by Stimulating the Face

We now summarize research that conveys direction by providing haptic stimuli to the face. For example, a method has been proposed that provides directional cues to users by generating a pulling sensation via a head-mounted device [4]. In addition, a navigation method that applies haptic stimuli to the cheeks has been proposed [8], but the main device also needs to be attached to the head, making it less convenient to wear. Another reported approach involves using airborne ultrasound to noninvasively relay information to the lips [3], but because this method requires placing a device externally, it is difficult to use as a wearable device.

Among the possible regions on the face for presenting information, the nose is another candidate. It has been found that, with respect to pressure, the nose has the highest sensitivity in the body—on par with the lips and cheeks—and that its sensitivity to vibration is also quite high, next only to that of the fingers, lips, and cheeks [14]. Thus, it may be possible to convey sufficient information, such as direction, even with weak stimuli. Furthermore, considering that many people wear glasses on a daily basis, it may be easier to attach a device to the nose compared to the lips or cheeks.

One example of a sensory presentation method targeting the nose is a technique in which the trigeminal nerve in the nasal septum

¹<https://www.northsense.com/>

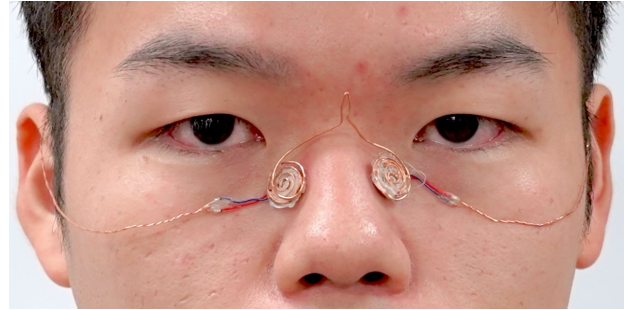


Figure 2: Prototype used in experiments

is electrically stimulated when presenting odors, allowing users to perceive the direction and intensity of scents [1]. However, there has been relatively little investigation into the use of tactile stimuli to the nose for presenting directional cues, suggesting room for further research. Hence, this study focuses on the nose to develop a device that is easy to put on or take off and remains comfortable to wear over extended periods.

3 NOSEKNOWSNORTH

3.1 Concept

We propose a method for indicating direction by applying vibratory stimuli to the nose. This idea was inspired by certain fish (such as eels) that possess magnetic receptors in their nasal cavities. Humans lack the ability to perceive geomagnetic fields consciously[13]; therefore, this study attempts to utilize tactile sensation as a means to acquire a sense of direction. Another reason is that tactile stimulation of the nose has not been extensively investigated thus far, and the high tactile sensitivity of the nose suggests that it is well-suited to providing information through haptic stimuli. Although the fingers are also considered highly sensitive body parts, they have a high degree of freedom of movement; in daily life, the positional relationship between the fingers and the face or torso changes significantly, making them unsuitable for receiving direction-indicating haptic stimuli. On the other hand, among facial areas, we focused on the nose because of eyeglasses. Eyeglasses are supported by the nose and ears, and people commonly wear them for extended periods on a daily basis, suggesting that mounting a device on the nose would impose little burden. Moreover, it is relatively easy to integrate a device into the nose pads of eyeglasses, and such a device would be unobtrusive, making it easy to use in everyday life. As a result, users may come to wear the device without being conscious of it, leading to a sense of augmented sensation as if they had gained a new sensory organ. For instance, by incorporating this device into smart glasses, it could be applied for navigation or notification functions (Figure 1(a)).

3.2 Implementation

Our device consists of a 9-axis sensor (accelerometer, gyroscope, magnetometer; BNO085 by CEVA), a microcontroller board for control (XIAO ESP32C6 by Seeed Studio), two coin-type vibration motors (left and right), a battery (Figure 1). In the experiment, the device was used before it was integrated into smart glasses. The user attaches the vibration motors at the same positions where the nose pads of eyeglasses normally rest on the nose (Figure 2). In this configuration, the vibration motors sit on the nose as if pinching it, without using any adhesive tape. The rated voltage of the vibration motors is 3 [V], and in the maximum vibration condition we applied 2 [V].

For instance, in an application that presents the direction of north, the system calculates the azimuth angle with respect to north

(0°) based on data from the 9-axis sensor, and outputs PWM signals to the two vibration motors accordingly to provide tactile feedback. Vibration is generated within a range of -90° to $+90^\circ$ relative to north. Vibration intensity is varied linearly depending on the angle. For example, when the azimuth angle is 0° , both the left and right motors vibrate at maximum intensity. When the angle is 90° , only one motor (either left or right) vibrates at maximum intensity, while the other is inactive. If the user is facing 45° to the right, the left motor vibrates at maximum intensity, and the right motor vibrates at half intensity. By combining various vibration patterns of the two motors, it is possible to indicate left, right, or the target direction. During the experiment, the device was connected to a PC via a wired connection for control.

4 EVALUATIONS

We conducted two experiments to confirm whether it is possible to present directions by tactile stimulation with the nose. In both experiments, participants were not told whether the identification results were correct or incorrect. The experiment participants consisted of three male individuals aged 23 to 24, who were invited to participate after providing their consent. A questionnaire was administered after the experiments. Participants rated the ease of discrimination between right and left, the comfort of the stimuli, and the comfort of wearing the glasses on a 5-point scale, and responded to questions about points of concern in a free-writing form. The study was not ethically reviewed because it was a simple and exploratory study, and the authors obtained comments from developers who are familiar with HCI.

4.1 User Study 1: Discrimination of Left and Right

When the device is mounted on the lateral sides of the nasal bridge, there was a concern that the vibration might propagate such that it would be unclear whether the left or right vibration motor was active. They were seated while wearing the device, and after the stimulus was presented, they were asked to report which of the left or right vibration motors was active. If they could not determine which side was vibrating, they answered “don’t know.” The order of left and right vibration stimuli was presented randomly, with each side presented five times (total of ten stimuli) per session. After a brief practice session, two sessions were performed, and the results from these two sessions were treated as the experimental data.

4.2 Results

In the experiment on left-right discrimination, responses of “don’t know” were recorded twice from one participant; aside from those two cases, the correct answer rate was 100%. Regarding the two “don’t know” responses, there is a possibility that the motor did not vibrate because it was driven under its rated voltage. From these results, it was confirmed that two-point discrimination is possible with the current vibration stimulus. In Weinstein et al.’s experiment [14], the two-point discrimination threshold on the nose is about 8 mm. Hence, when the device is attached to the lateral sides of the nasal bridge, it is considered feasible to discriminate left and right vibrations.

4.3 User Study 2: Discrimination of Eight Directions

Next, we investigated whether it is possible to discriminate eight directions using this device. Eight directional markers labeled 1 through 8 at 45° intervals were placed on the circumference of a circle with a radius of 50 cm centered at the participant’s feet. Using data from a 9-axis sensor, the stimulus was presented so that the vibration intensity would be strongest when the participant was facing the target direction.

While standing in place and changing their facing direction, participants were asked to judge which direction was being presented and then indicate their answer by pointing to the marker on the floor.

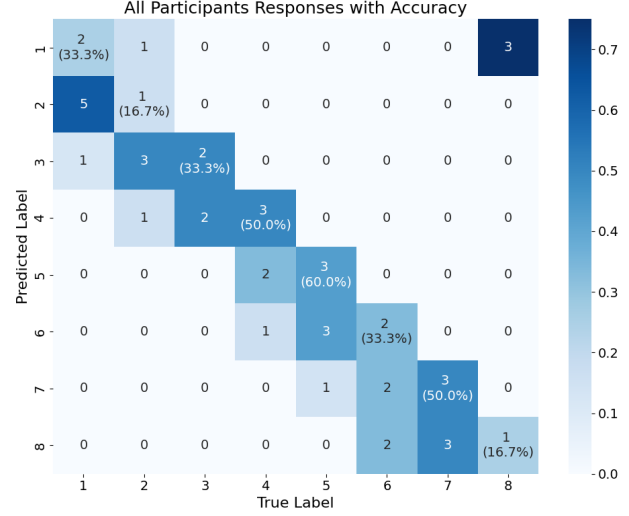


Figure 3: Responses in the direction-discrimination experiment by all participants. Labels 1 to 8 on the axis represent the numbers clockwise in 8 directions divided at 45° intervals

Each of the eight directions was presented twice, for a total of 16 presentations, in random order. After a brief practice session, the aforementioned 16 presentations were carried out, and these results were used for analysis.

4.4 Results

We describe the results of an experiment to determine whether participants could discriminate among eight directions. Figure 3 shows the responses of all participants ($N = 3$). The vertical axis represents the direction label reported by the participants, while the horizontal axis represents the correct direction label (numbered 1 to 8 at 45° intervals clockwise), and the cell values show the number of responses (up to two responses per participant, and with three participants, the maximum is six). Values in parentheses indicate accuracy rates. The overall accuracy rate, taking the simple average by label, was about 36%. Individually, the minimum accuracy was 16.7% (Labels 2 and 8), and the highest accuracy was 60% (Label 5).

4.4.1 Analysis of Error Distribution

To further evaluate accuracy for each direction, we computed the angular difference between the reported direction and the correct direction and calculated the mean error relative to the correct direction. We then analyzed the distribution of angular errors between the reported and correct directions. Figure 4 shows the distribution of each error and its number. The mean angular error was -32.55° , indicating that the reported directions tended to be skewed counter-clockwise relative to the correct directions overall. Moreover, the standard deviation was 32.10° , indicating large variance in the errors. Next, two main peaks in the error distribution were observed near 0° and -45° , suggesting a high frequency of confusion with adjacent directions (i.e., $\pm 45^\circ$).

From these results, it can be inferred that while the device has a certain level of accuracy for direction presentation, errors frequently occur in adjacent directions.

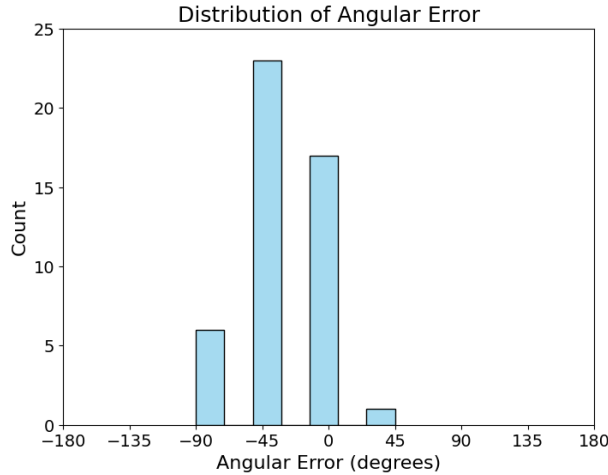


Figure 4: Distribution of angular errors between the reported and correct directions

Horizontal axis: angular error, Vertical axis: the number of errors

4.4.2 Calculation of Accuracy Considering Error and Resolution

Based on the analysis of the error distribution and the errors for each direction, we examined how the accuracy would change if we allowed for an angular deviation of -45° (one label increment). First, we determined what percentage of the data could be explained by allowing a deviation of -45° . Specifically, we calculated the percentage of data for the 0° and -45° labels in Figure 4 and found that it could explain 85.11 % of the total data. This suggests that the resolution of direction presentation in this system is about 45° .

Next, we analyzed the percentage of correct responses in each direction when a -45° deviation was allowed. Figure 5 represents the percentage of responses in each direction. The vertical axis represents the percentage of responses, and the horizontal axis represents the direction labels that were answered. In this graph, blue represents the percentage of correct responses in a certain direction, red represents the percentage of responses that deviated -45° from the correct answer, and gray represents the percentage of other responses. The average percentage of correct answers was about 36%, and the average percentage of answers that were off by -45° was about 49%.

This means that the overall percentage of correct responses was 85.11% if the responses that were off by -45° were included in the correct responses, which is higher than the overall percentage of correct responses of 36% when no error was allowed. When a $\pm 45^\circ$ deviation was allowed, the overall correct response rate was 87.23%.

These results suggest that the device has a resolution of within 45° and demonstrates high direction-indication ability when some margin of error is permitted. However, these are the results under the current stimulus-presentation method, and the small number of participants as well as potential implementation issues with the experimental device should be taken into consideration.

4.5 User Feedback and Discussion

From the post-experiment questionnaire, we received feedback on the vibration presentation method, suggesting that weakening the vibration as one approaches the correct direction, and not vibrating when precisely on the correct direction, might make it easier to understand. There were also comments such as “my nose felt

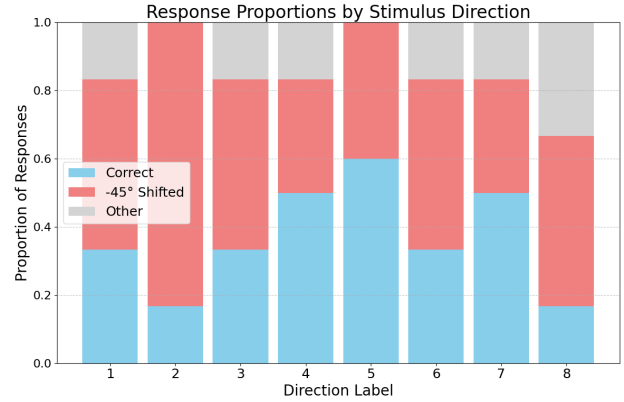


Figure 5: Accuracy for each direction when allowing an error from 0° to -45°

Horizontal axis: direction label, Vertical axis: accuracy

slightly itchy after prolonged use” and reports of a sense of pressure. In this experiment, the vibration stimulus became stronger as one approached the correct direction, and it was continuously presented throughout the session, which may have caused discomfort to participants. Thus, there is room to refine how the vibration is presented depending on the direction faced, for example by adding a rhythm to the vibration to avoid continuous stimulation. We also received the comment, “the feeling changed quite a bit depending on how the device contacted the nose.” In particular, the current prototype merely pinches the nose, and its position may shift due to the vibration during use. To improve the ease of perceiving the stimulus as well as wearing comfort, it is necessary to explore how best to position the device and create closer contact with the nose, thereby identifying the optimal site for stimulus presentation.

5 CONCLUSION

In this study, we proposed NoseKnowsNorth, a device that provides directional cueing through vibration stimuli applied to the nose. This device can be integrated into the nose pads of smart glasses, leveraging the nose’s high tactile sensitivity to offer intuitive and unobtrusive directional cues. We conducted experiments using the developed prototype to evaluate its directional indication capability and confirmed that participants could nearly perfectly distinguish between left and right when vibrations were applied to the nose. In addition, we confirmed that the prototype was able to present directions with a resolution of within 45° in 8-directional guidance.

Moving forward, we plan to enhance the device’s comfort and usability by refining its design and exploring more effective vibration patterns for directional guidance. Furthermore, we aim to investigate whether wearing the device for extended periods could enable users to acquire a sense of direction, testing hypotheses about augmenting bodily sensations through the device. This exploration will further examine the potential for sensory augmentation facilitated by this technology.

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REFERENCES

- [1] J. Brooks, S.-Y. Teng, J. Wen, R. Nith, J. Nishida, and P. Lopes. Stereo-smell via electrical trigeminal stimulation. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*,

- CHI '21. Association for Computing Machinery, New York, NY, USA, 2021. 2
- [2] R. Gibb, B. Ercoline, and L. Scharff. Spatial disorientation: Decades of pilot fatalities. *Aviation, Space, and Environmental Medicine*, 82(7):717–724, 2011. 1
 - [3] A. Jingu, T. Kamigaki, M. Fujiwara, Y. Makino, and H. Shinoda. Lip-notif: Use of lips as a non-contact tactile notification interface based on ultrasonic tactile presentation. In *The 34th Annual ACM Symposium on User Interface Software and Technology*, UIST '21, p. 13–23. Association for Computing Machinery, New York, NY, USA, 2021. 2
 - [4] W. Li, T. Nakamura, and J. Rekimoto. Remoconhanger: Making head rotation in remote person using the hanger reflex. In *Adjunct Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology*, UIST '22 Adjunct. Association for Computing Machinery, New York, NY, USA, 2022. 2
 - [5] A. Matsuda, K. Nozawa, K. Takata, A. Izumihara, and J. Rekimoto. Hapticpointer: A neck-worn device that presents direction by vibrotactile feedback for remote collaboration tasks. In *Proceedings of the Augmented Humans International Conference*, AHs '20. Association for Computing Machinery, New York, NY, USA, 2020. 2
 - [6] Y. Mizohata, N. Ohnui, A. Kikukawa, A. Kobayashi, M. Fujita, K. Takada, S. Tachibana, T. Iwamoto, D. Yamaguchi, M. Kimura, and S. Bekku. Spatial disorientation and countermeasure. *Aeromedical Laboratory Reports*, 56(4):79–93, 2016. 2
 - [7] S. Nagel, C. Carl, T. Kringe, R. Märtin, and P. König. Beyond sensory substitution – learning the sixth sense. *Journal of neural engineering*, 2:R13–26, 01 2006. 2
 - [8] F. Nakamura, A. Verhulst, K. Sakurada, and M. Sugimoto. Virtual whiskers: Spatial directional guidance using cheek haptic stimulation in a virtual environment. In *Proceedings of the Augmented Humans International Conference 2021*, AHs '21, p. 141–151. Association for Computing Machinery, New York, NY, USA, 2021. 2
 - [9] R. L. Peiris, Y.-L. Feng, L. Chan, and K. Minamizawa. Thermal-bracelet: Exploring thermal haptic feedback around the wrist. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, CHI '19, p. 1–11. Association for Computing Machinery, New York, NY, USA, 2019. 2
 - [10] S. Schaack, G. Chernyshov, K. Ragozin, B. Tag, R. Peiris, and K. Kunze. Haptic collar: Vibrotactile feedback around the neck for guidance applications. In *Proceedings of the 10th Augmented Human International Conference 2019*, AH2019. Association for Computing Machinery, New York, NY, USA, 2019. 2
 - [11] Y. Tanaka, J. Nishida, and P. Lopes. Electrical head actuation: Enabling interactive systems to directly manipulate head orientation. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*, CHI '22. Association for Computing Machinery, New York, NY, USA, 2022. 2
 - [12] K. Tsukada and M. Yasumura. Activebelt: Belt-type wearable tactile display for directional navigation. In N. Davies, E. D. Mynatt, and I. Siio, eds., *UbiComp 2004: Ubiquitous Computing*, pp. 384–399. Springer Berlin Heidelberg, Berlin, Heidelberg, 2004. 2
 - [13] C. X. Wang, I. A. Hilburn, D.-A. Wu, Y. Mizuhara, C. P. Cousté, J. N. H. Abrahams, S. E. Bernstein, A. Matani, S. Shimojo, and J. L. Kirschvink. Transduction of the geomagnetic field as evidenced from alpha-band activity in the human brain. *eNeuro*, 6(2), 2019. 2
 - [14] S. Weinstein. Intensive and extensive aspects of tactile sensitivity as a function of body part, sex and laterality. *The skin senses*, pp. 223–261, 1968. 2, 3